

CampusBridge

A Career Intelligence and Alumni Engagement Platform

Project Report

Prepared for Surana College

This report describes the design and implementation of CampusBridge, a MERN based alumni management, referral, expert talk, and FDP coordination platform for a college environment.

1. Introduction

CampusBridge is a web based platform designed for Surana College to maintain alumni records and use those records for practical institutional engagement. The application supports annual alumni data collection, department based alumni search, student referral coordination, and Expert Talk or Faculty Development Program invitations. The goal is not only to store alumni information, but also to convert alumni relationships into useful academic and career support activities.

The system follows a department-admin model. After login, an admin can view only the alumni and requests belonging to the admin's registered department. This keeps the experience simple for staff members and avoids exposing alumni data across unrelated departments.

2. Problem Statement

Colleges often collect alumni details through spreadsheets, messaging groups, or disconnected forms. These approaches become difficult to maintain over time. Departments may not know which alumni are available for expert talks, FDP support, referrals, mentoring, or industry interaction. Tracking whether an alumnus accepted, declined, viewed, or responded to a request is also difficult without a structured workflow.

CampusBridge solves this by providing a centralized, department scoped alumni management system with secure response links for alumni. Alumni do not need to create an account for every request. Instead, the college can send a private link where the alumnus can respond directly.

3. Objectives

- Collect yearly alumni information through a public registration form.
- Allow department admins to search only their department alumni after login.
- Support student referral requests with resume upload and secure alumni response links.
- Support Expert Talk and FDP requests with secure alumni accept, decline, and more-details responses.
- Track email delivery, alumni view time, response time, and request status.
- Provide a clean dashboard suitable for real college administrative use.

4. Stakeholders

Stakeholder	Role in System
Department Admin	Logs in, searches alumni, creates referral and Expert/FDP requests, tracks responses.
Alumni	Registers details and responds to college requests through secure no-login links.
Current Student	Receives referral support through college initiated alumni connections.
Faculty / Department	Uses alumni network for expert talks, FDPs, workshops, and mentoring.
College Management	Benefits from organized alumni engagement and measurable outreach.

5. Technology Stack

Layer	Technology	Purpose
Frontend	React 19 with Vite	Single page user interface for admin dashboard and public forms.

Layer	Technology	Purpose
Backend	Node.js and Express	REST APIs, authentication, email workflow, file upload handling.
Database	MongoDB with Mongoose	Stores alumni, admin users, referrals, and Expert/FDP requests.
Email	Nodemailer SMTP	Sends automatic request emails to alumni.
Upload	Multer	Handles PDF, DOC, and DOCX resume uploads for referral requests.

6. Implemented Modules

6.1 Alumni Registration

The public alumni form allows passed-out students to submit their details, including department, batch, current company or institution, role, skills, sector, contact preference, availability for expert talks, availability for FDP, and expertise topics. This forms the base data set used by departments.

6.2 Admin Login and Department Based Access

Admin users are seeded for Surana College departments. Passwords are stored as PBKDF2 hashes. After login, the backend returns a signed admin session token. Protected APIs derive department scope from the authenticated admin instead of trusting a department selected in the browser.

6.3 Alumni Search Dashboard

The dashboard supports search by name, company, skills, topics, sector, batch, expert talk availability, and FDP availability. Results are limited to the logged in admin's department. Each alumni row provides actions to connect a student, create an Expert Talk request, or create an FDP request.

6.4 Referral Request Module

The referral workflow lets the department connect a current student with an alumnus. The admin enters student details, target company, target role, skills, request reason, and an optional resume file. The system generates a private response token and attempts to email the alumnus automatically.

6.5 Expert Talk / FDP Request Module

This module allows admins to invite alumni for Expert Talk or FDP sessions. The request includes session title, audience, proposed date, duration, mode, topic, and request details. Alumni can respond using a secure link without creating an account.

6.6 Request Trackers

Referral and Expert/FDP trackers use compact cards. Each card shows summary information by default and expands on click to show full details, private response link, email actions, status update controls, and response timeline.

7. System Architecture

CampusBridge follows a client-server architecture. The React frontend communicates with Express REST APIs. Express validates requests, enforces admin authorization, performs database operations through Mongoose, and triggers email delivery through Nodemailer. MongoDB stores all persistent data.

Component	Responsibility
React UI	Admin login, dashboard, search, forms, trackers, public response pages.

Component	Responsibility
Express API	Routes for authentication, alumni, referrals, Expert/FDP requests, and public responses.
Mongoose Models	Structured schemas for Alumni, AdminUser, ReferralRequest, and ExpertRequest.
Email Service	Builds and sends SMTP emails containing secure private response links.
MongoDB	Stores college data, statuses, timestamps, and request tokens.

8. Database Design

Entity	Important Fields
AdminUser	fullName, email, passwordHash, department, role, status
Alumni	fullName, email, phone, graduationYear, department, companyName, jobRole, skills, expertiseTopics, availability flags
ReferralRequest	student details, target company, resume metadata, alumni reference, responseToken, status, timestamps
ExpertRequest	requestType, title, department, audience, proposedDate, duration, mode, topic, alumni reference, responseToken, status

The design uses references from requests to alumni records. This avoids duplication while still allowing request tracking. Indexes on department and status improve dashboard filtering. Response tokens are unique and are used for no-login public response pages.

9. Key Workflows

9.1 Alumni Registration Workflow

- College shares the public alumni form link.
- Alumnus submits current details.
- Record is saved in MongoDB.
- Department admin can search the alumnus after login.

9.2 Referral Workflow

- Admin searches alumni.
- Admin selects an alumnus and creates referral request.
- System generates secure response token.
- System sends email if SMTP is configured.
- Alumnus accepts, declines, asks for more details, or marks student contacted.
- Admin tracks status and timestamps.

9.3 Expert/FDP Workflow

- Admin selects alumni for Expert Talk or FDP.
- Admin fills session title, audience, date, duration, mode, and topic.
- System creates private response link and sends email.

- Alumnus responds without login.
- Admin tracks request status and schedules session if accepted.

10. API Overview

API Area	Endpoints
Authentication	POST /api/auth/login, GET /api/auth/me
Alumni	POST /api/alumni, GET /api/alumni, GET/PUT/DELETE /api/alumni/:id
Referrals	POST /api/referrals, GET /api/referrals, PATCH /api/referrals/:id, POST /api/referrals/:id/send-email
Referral Public	GET /api/referrals/public/:id, PATCH /api/referrals/public/:id/respond, GET /api/referrals/public/:id/resume
Expert/FDP	POST /api/expert-requests, GET /api/expert-requests, PATCH /api/expert-requests/:id, POST /api/expert-requests/:id/send-email
Expert/FDP Public	GET /api/expert-requests/public/:id, PATCH /api/expert-requests/public/:id/respond

11. Security Considerations

- Admin APIs require a signed token and derive department scope from the authenticated admin.
- Passwords are stored as salted PBKDF2 hashes rather than plain text.
- Alumni response links use unique private response tokens.
- Referral resume downloads require the correct token.
- File uploads are restricted to PDF, DOC, and DOCX with a 5 MB limit.
- Production deployment should use HTTPS, strong ADMIN_SESSION_SECRET, secure cookies, rate limiting, and audited SMTP credentials.

12. Testing and Validation

The project was validated using production builds and backend smoke tests. The Vite build verifies frontend compilation. Backend syntax checks validate new route, service, and model files. API checks verified authenticated access and department scoped data retrieval.

Test Area	Validation
Frontend Build	npm run build completed successfully.
Admin Auth	CSE admin login returns a signed token and department details.
Department Scoping	CSE admin receives only Computer Science records.
Protected API	Unauthenticated alumni API returns 401.
Expert/FDP API	Authenticated expert request listing endpoint responds successfully.

13. Limitations

- The current login stores the admin session in browser storage; production should prefer httpOnly secure cookies.

- There is no super-admin screen yet for managing departments and admin accounts.
- Email delivery depends on SMTP configuration and is not retried through a queue.
- The application currently uses simple REST APIs without automated test coverage.
- Advanced analytics and reporting dashboards are not yet implemented.

14. Future Scope

- Add super-admin controls for departments, admins, and institution level reports.
- Add CSV/Excel import and export for alumni records.
- Add event calendar integration for accepted Expert Talk and FDP sessions.
- Add automatic reminders for pending alumni responses.
- Add analytics for alumni engagement rate, accepted requests, sessions completed, and department activity.
- Add role based access control, audit logs, and production-grade authentication.
- Add AI assisted alumni matching based on skills, company, topic, and past response behavior.

15. Conclusion

CampusBridge provides a practical, deployable foundation for alumni engagement at Surana College. It moves alumni management away from manual spreadsheets and towards a structured system where departments can search alumni, request referrals, invite experts, track responses, and maintain institutional memory. With further hardening and analytics, the platform can grow into a complete career intelligence and alumni engagement solution.